

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019-2020 (Even Semester)

Innovative Practice Description for Unit-I

Degree, Semester & Branch: IV Semester B.E. ECE B

Course Code & Title: EC8452 Electronic Circuits II

Name of the Faculty member: Mr.A.Azhagu Jaisudhan Pazhani

Name of the Topic: Effect of feedback on input impedance, Analysis of series-series feedback amplifiers and Stability problem- Gain and Phase margins

Name of the Innovative Practices: Sketch noting, One minute paper and Quescussion

Date and Duration: 02.01.2020 (15 Minutes), 06.01.2020 (10 Minutes) and 11.01.2020 (10 Minutes)

Description:

(i) Sketch noting:

- The activity sketch noting is a purposeful tool while listening to something interesting.
- Sketch notes don't require high drawing skills, but do require a skill to visually synthesize and summarize via shapes, connectors, and text. Sketchnotes are as much a method of note taking as they are a form of creative expression

(ii) One Minute Paper:

This method used at the end of class for assessing student performance. Advantage of this technique is that it provides quick feedback on whether the concept is reached to the students. Additionally, by asking students to add a question at the end, this assessment becomes an integrative task. Sometimes, instead of asking for the main point, a professor may wish to probe for the most disturbing or most surprising item.

(iii) Quescussion:

Quescussion is a type of discussion that is conducted entirely in the form of questions. It was developed by Professor Paul Bidwell of the English Department at the University of Saskatchewan where he has used it with great success in the teaching of poetry. It has proved to be very useful in handling a variety of subjects even, or perhaps especially, very controversial ones, and works across a wide range of class sizes.

Goals (Learning Outcomes):

- The students will be able to analyze the effect of feedback on input impedance.
- The students will be able to analyze the series-series feedback amplifier.
- The students will be able to explain about stability problems in feedback amplifier.

Use of appropriate method:

Justification for choosing the following Activities:

- **Sketch noting:** Students can able to draw and make connection between various circuit components.
- **Minute Paper:** Advantage of minute papers is that they provide manageable amounts of timely and useful feedback for a minimal investment of time and energy.
- **Quescussion:** Students can able to easily interact with neighbours in order share the ideas related to posing questions.

Effective Presentation: (Implementation (Plan & Execution) with Proof):

- **Sketch noting:** All the students are participated in this activity. Some of the students plot the diagram in the board.
- **Minute Paper:** Students used small paper to write a few points about topologies of feedback amplifier
- **Quescussion:** Students are easily understand the stability issues in feedback amplifiers

Sketch noting for Effect of feedback on input impedance on 02.01.2020 (15 Minutes)



One Minute Paper for Analysis of series-series feedback amplifiers on 06.01.2020 (10 Minutes)



**Quescussion for Stability problem- Gain and Phase margins
on 11.01.2020 (10 Minutes)**



Significance of Results: (Assessment of Effectiveness/Success of the Activity):

- **Sketch Noting:** With interest, most of the students interconnect the circuit component.
- **Minute Paper:** Helps the students to recall the important feedback topologies
- **Quescussion:** The success of this activity is evaluated through internal test questions.

Changes required for future (if required):

- Students requested to give topics before a day for their preparation for zero minute speech.
- Sometime need to discuss the concepts during quesquession session so more time has to allocate for this activity in the future.

Reflective Critique:

Challenges:

- **Sketch Noting:** During sketch noting few students hesitated to participate in the activity.
- **Minute Paper:** Students struggled to recollect the points
- **Quescussion:** Participants must wait until four (this number can vary with the size of the class) other people have spoken before they can speak again.

Benefits:

- **Sketch Noting:**
 - Students can able to understand how to design a circuit and connection between the various circuit components.
- **Animation:**
 - The animation is eye-catching, It is useful when quickly getting and holding the audience's attention.
 - The animation can show the imagined objects in the gesticulation, It is ideal for demonstrating processes.
- **Quescussion:**
 - In large classes, it allows many students to make brief contributions without interventions by the professor.
 - More question discussion for their better understanding of the topic

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	3	1	2	-	-	2	-	2	1	1

CO1: The students will be able to analyze the concepts of feedback Amplifiers in various applications.

References:

1. Robert L. Boylestad and Louis Nasheresky, "Electronic Devices and Circuit Theory", 10th Edition, Pearson Education / PHI, 2008
2. David A. Bell, "Electronic Devices and Circuits", Fifth Edition, Oxford University Press, 2008.
3. Millman J. and Taub H., "Pulse Digital and Switching Waveforms", TMH, 2000.
4. Millman and Halkias. C., Integrated Electronics, TMH, 2007.
5. <https://teachingcommons.lakeheadu.ca/qnescussions>